Sparse Gradient Aggregation

Anonymous

1 Setting

We minimise the average $\frac{1}{M}\sum_{m=1}^{M} F_m(x)$ of the local objectives over the workers. The dispersion term $\frac{1}{M}\sum_{m=1}^{M}\|\nabla F_m(x)-\nabla F(x)\|^2$ controls the aggregation error. The step size is set to $\eta=\frac{1}{L}\sqrt{\frac{\sigma^2}{T}}$ in every experiment we report.

Each worker m holds a local dataset and computes a stochastic gradient with bounded variance. The aggregation step averages the compressed residuals across the M workers at every round.

Assumption 1 (Smoothness)**.** *Each F_m is L-smooth, that is $\|\nabla F_m(x)-\nabla F_m(y)\|\le L\|x-y\|$ for all x.*

Assumption 2 (Bounded variance)**.** *The averaged squared deviation of the local gradients from the global gradient is bounded by a constant.*

2 Main result

Our main result bounds the residual error of the aggregation rule under the two assumptions above.

Theorem 1. *Under Assumptions 1 and 2, the iterates of the method satisfy the bound stated below.*

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\|\nabla F(x_t)\|^2\le\frac{c\,(F(x_0)-F^\star)}{\eta T}+c\,\eta L\sigma^2, \tag{1}$$

Here $c>0$ is an absolute constant that does not depend on T.

Lemma 1. *For all t larger than $t_0=\lceil 2/\delta\rceil$ we have the lower bound stated below.*

$$\mathbb{E}\|\bar{e}_t\|^2\ge\frac{\eta^2(1-\delta)}{4}\,\zeta^2. \tag{2}$$

The proof is deferred to Appendix A, and the constants are not optimised.

3 Discussion

The bound degrades gracefully as the compression ratio grows towards one half of the coordinates. Collecting the surviving terms gives the claimed lower bound on the expected residual norm. We expect the same argument to apply to the momentum variant without further modification.